

11. Project Planning

Project Planning Requirements

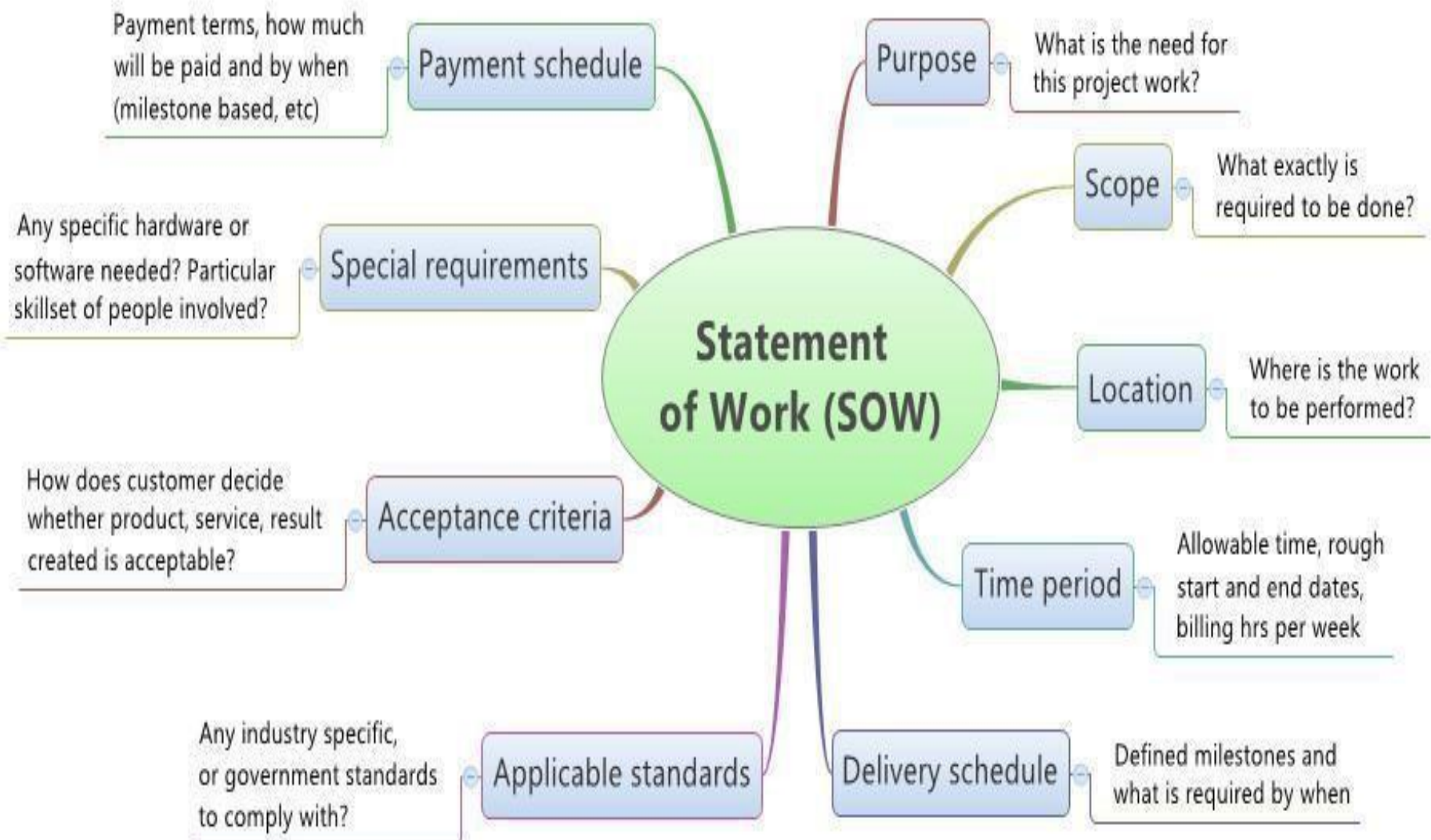
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- Statement of Work (SOW)
- Project Specifications
- Milestone Schedules
- Work Breakdown Structure (WBS)
- Organizational Breakdown Structure (OBS)
- Responsibility Assignment Matrix (RAM)

Statement of Work

- A narrative description of work to be accomplished
- It usually includes;
 - › Objectives of project
 - › A brief description of work
 - › Constraints, if any
 - › Specifications
 - › Schedules





Project Specifications

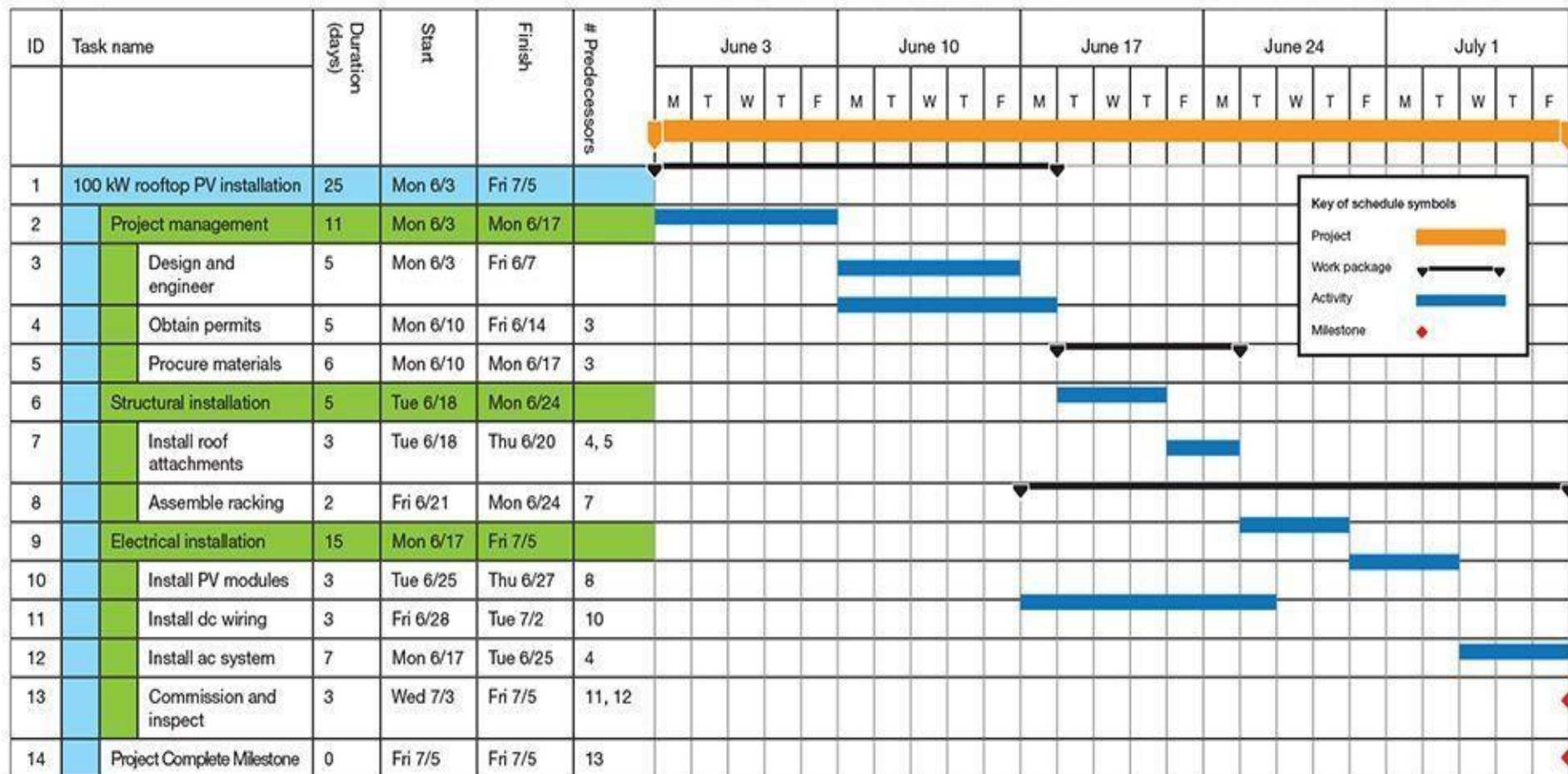
- A list separately identified or called out as part of the statement of work.
- Used for man-hour, equipment, and material estimates.
- Small changes in a specification can cause large cost overruns.



Milestone Schedules

- Contains information as:
 - > Project start date
 - > Project end date
- Other major milestones
 - > (review meetings, prototype available, procurement, testing)
- Data items
 - > (deliverables or reports)

A Gantt chart is the most commonly used scheduling tool in the construction industry. Here is a sample Gantt chart for a 100 kW rooftop PV installation.



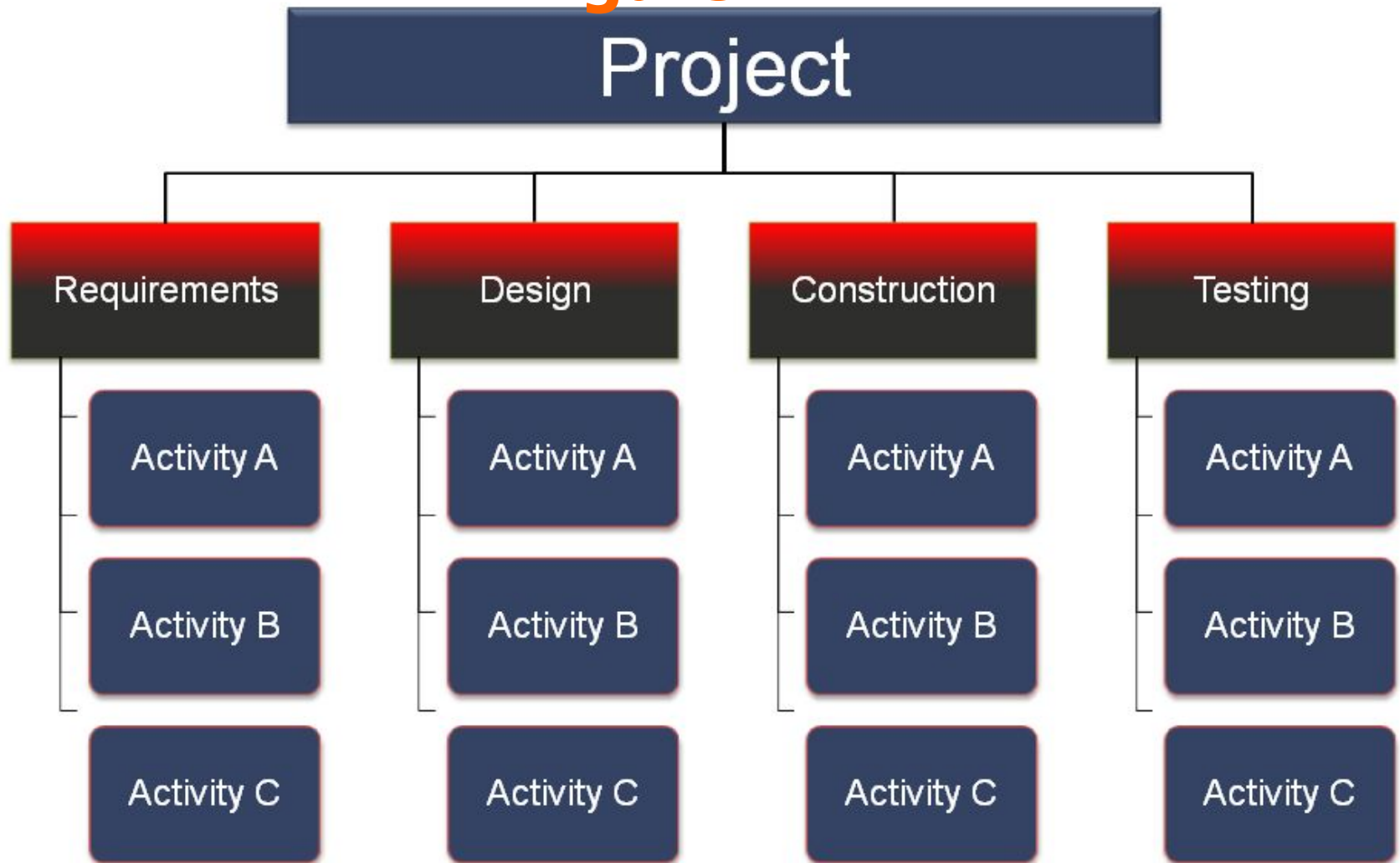
Work Breakdown Structure

- A deliverable-oriented hierarchical decomposition of the work to be executed by the project team to accomplish the project objectives and create the required deliverables.
- A Work Breakdown Structure (WBS) is a hierarchical structure of things that the project will make or outcomes that it will deliver.
- A work breakdown structure defines all the things a project needs to accomplish, organized into multiple levels, and displayed graphically.
- It tells what needs to be done.

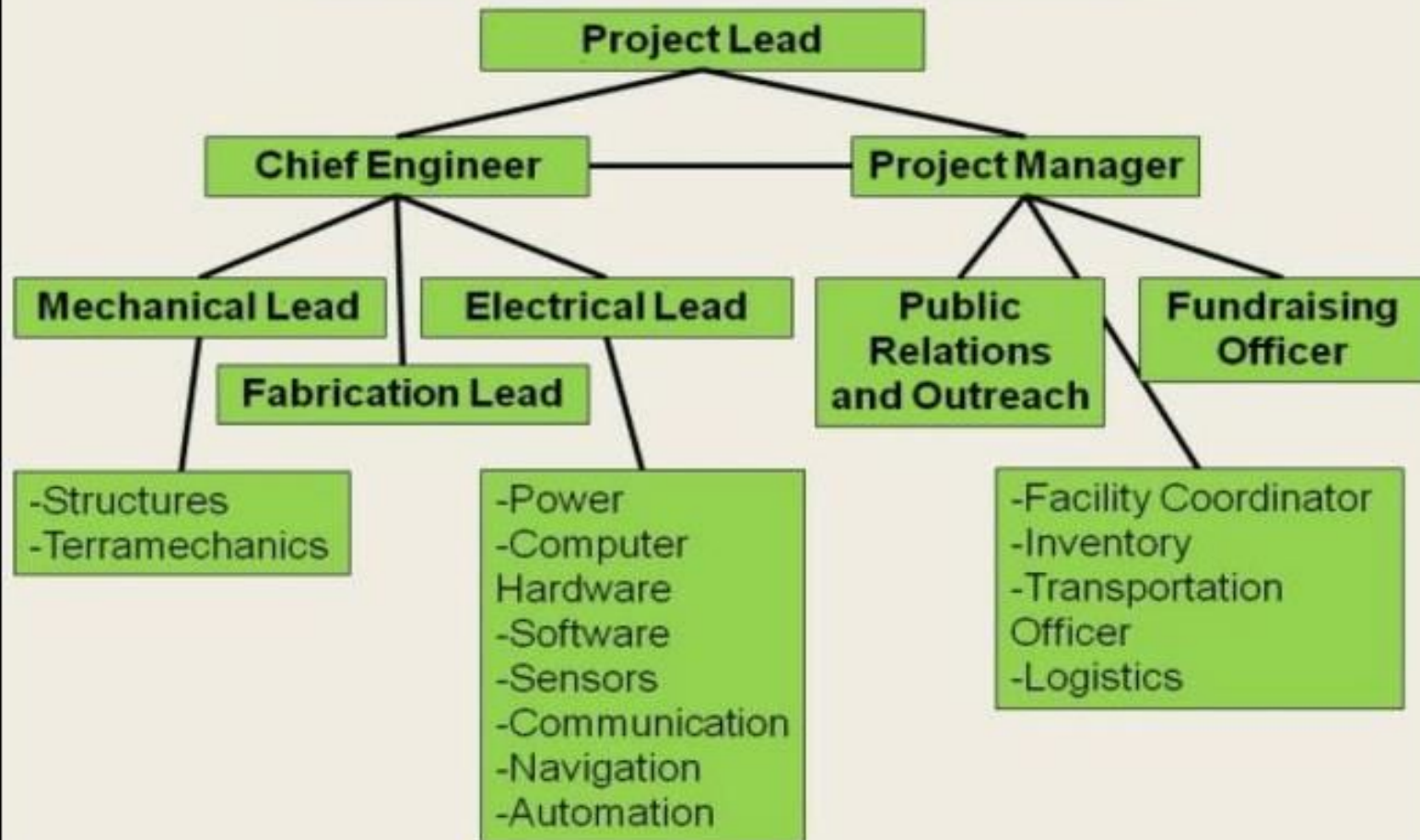
WBS

Figure

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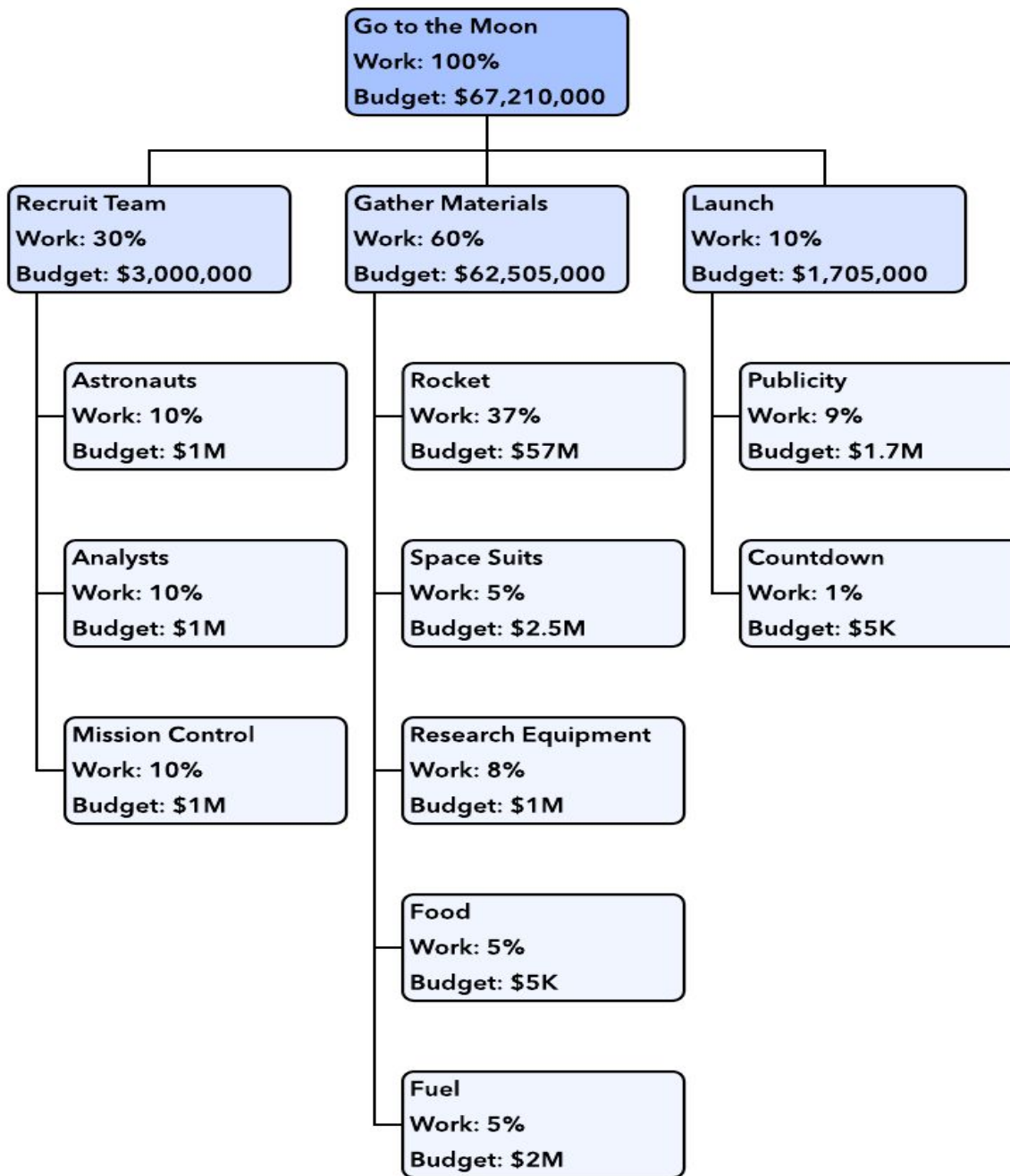


Work Breakdown Structure



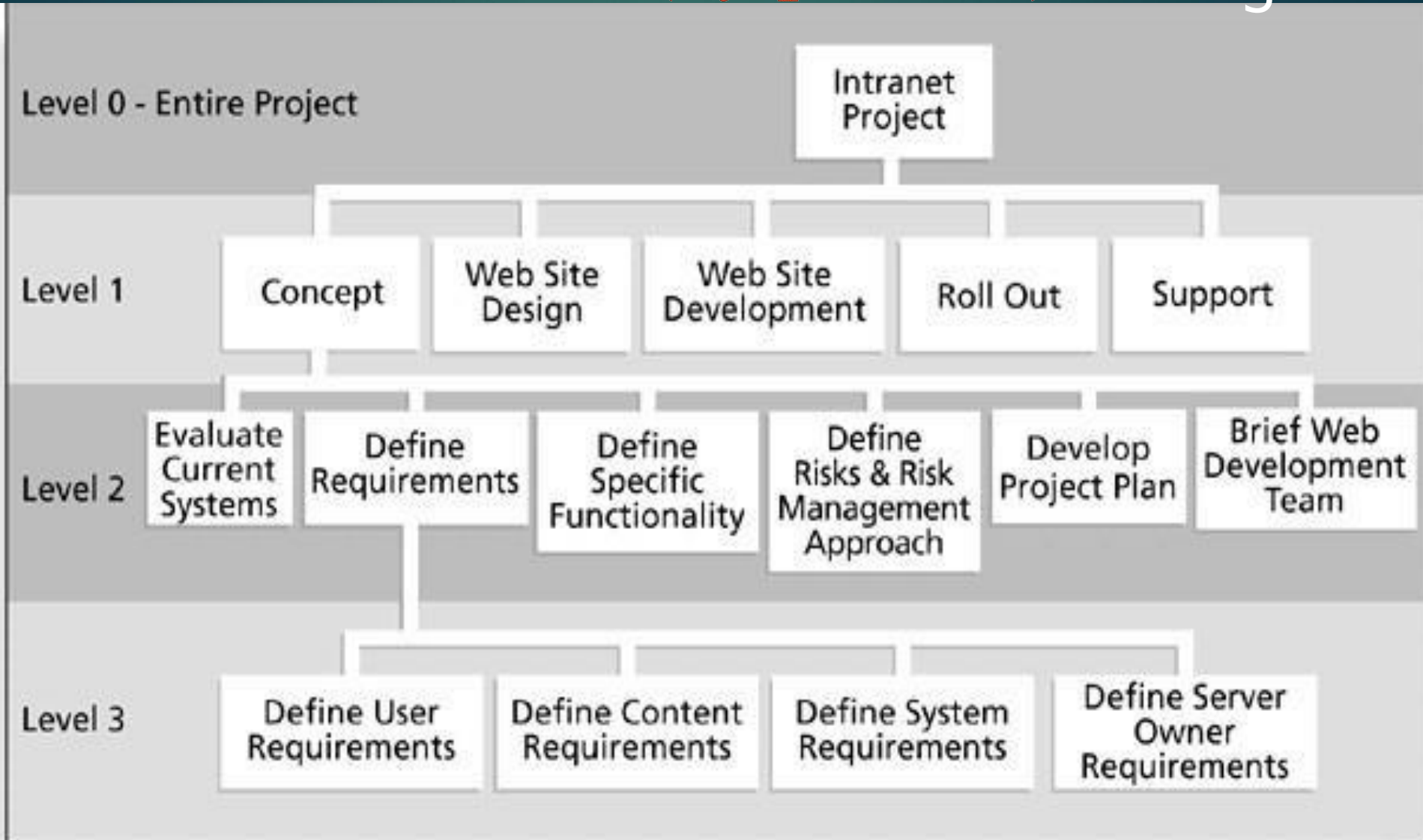
Need for WBS

- ① Three reasons to use a WBS in your projects.
- ② It helps more accurately and specifically define and organise the scope of the total project.
- ③ To help with assigning responsibilities, resource allocation, monitoring the project, and controlling the project.
- ④ It allows double check all the deliverables' specifics with the stakeholders and make sure there is nothing missing or overlapping.



Intranet WBS (by process)

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3



1.0

Concept

1. Evaluate current systems
2. Define Requirements
 1. Define user requirements
 2. Define content requirements
 3. Define system requirements
 4. Define server owner requirements
3. Define specific functionality
4. Define risks and risk management approach
5. Develop project plan
6. Brief web development team

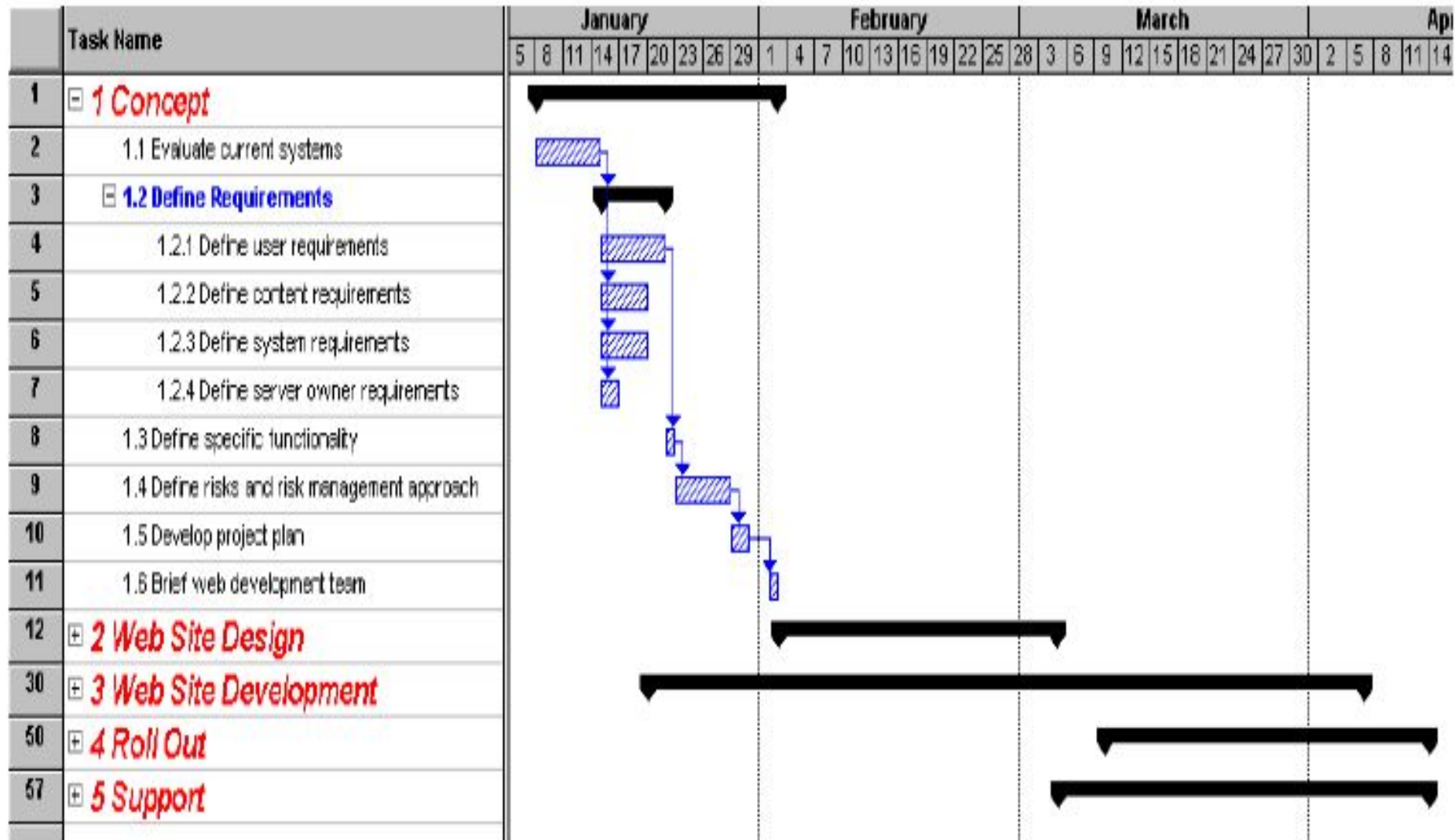
- ▶ 2.0 Web Site Design
- ▶ 3.0 Web Site Development
- ▶ 4.0 Roll Out
- ▶ 5.0 Support

Intranet Gantt Chart & WBS

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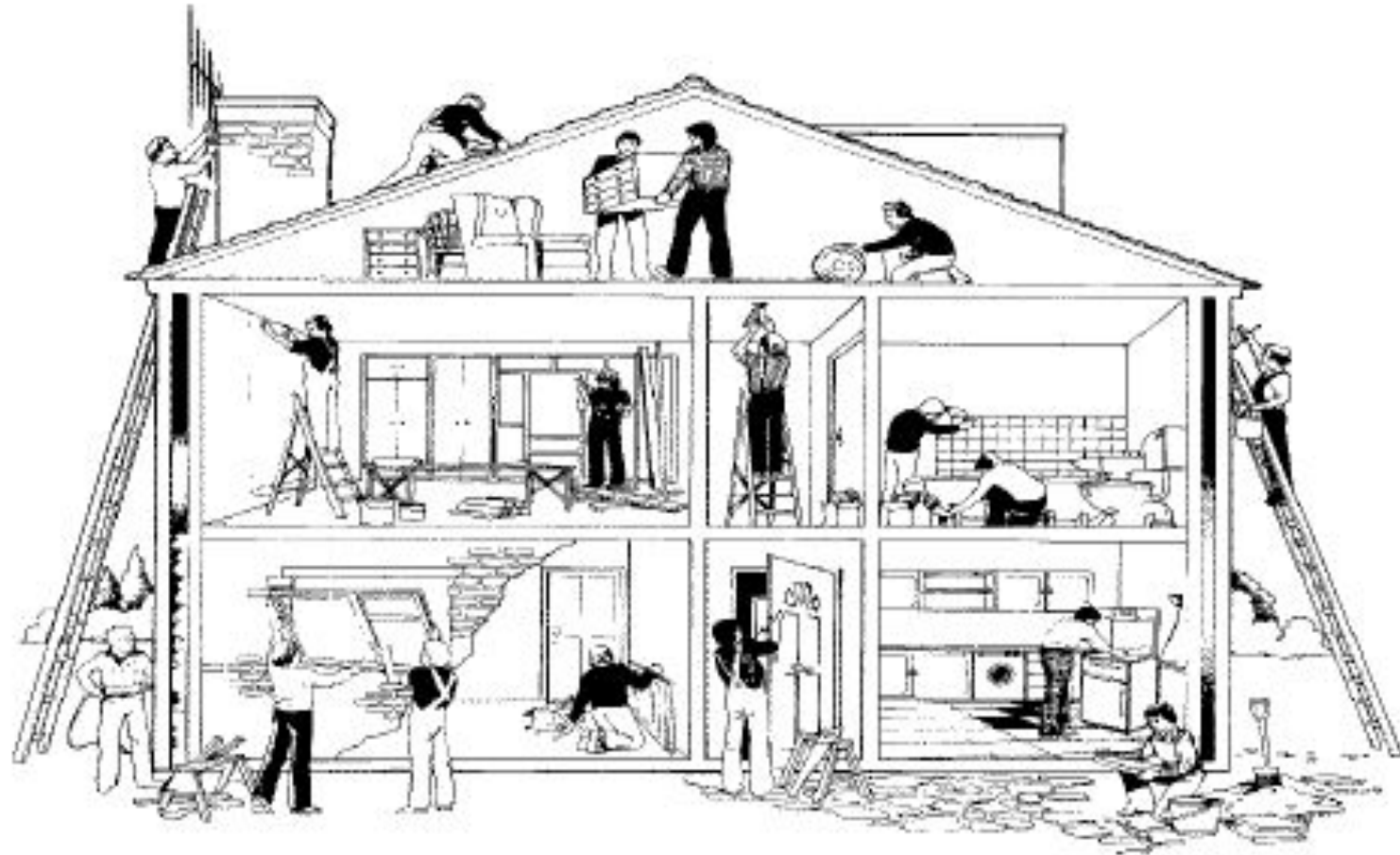
WBS

Gantt Charts



WBS Example – Construction

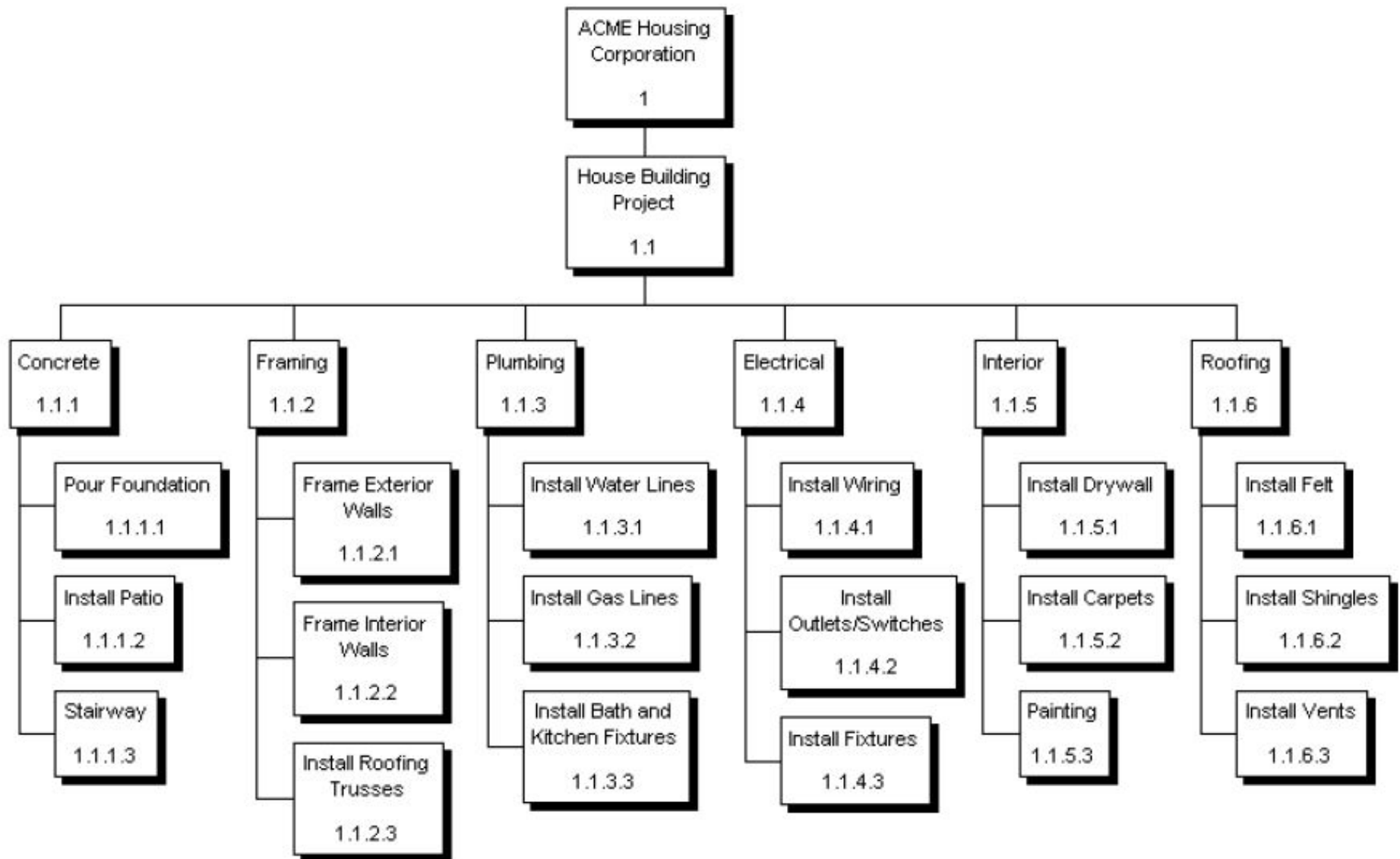
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Hierarchal

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Organizational Breakdown Structure (OBS)

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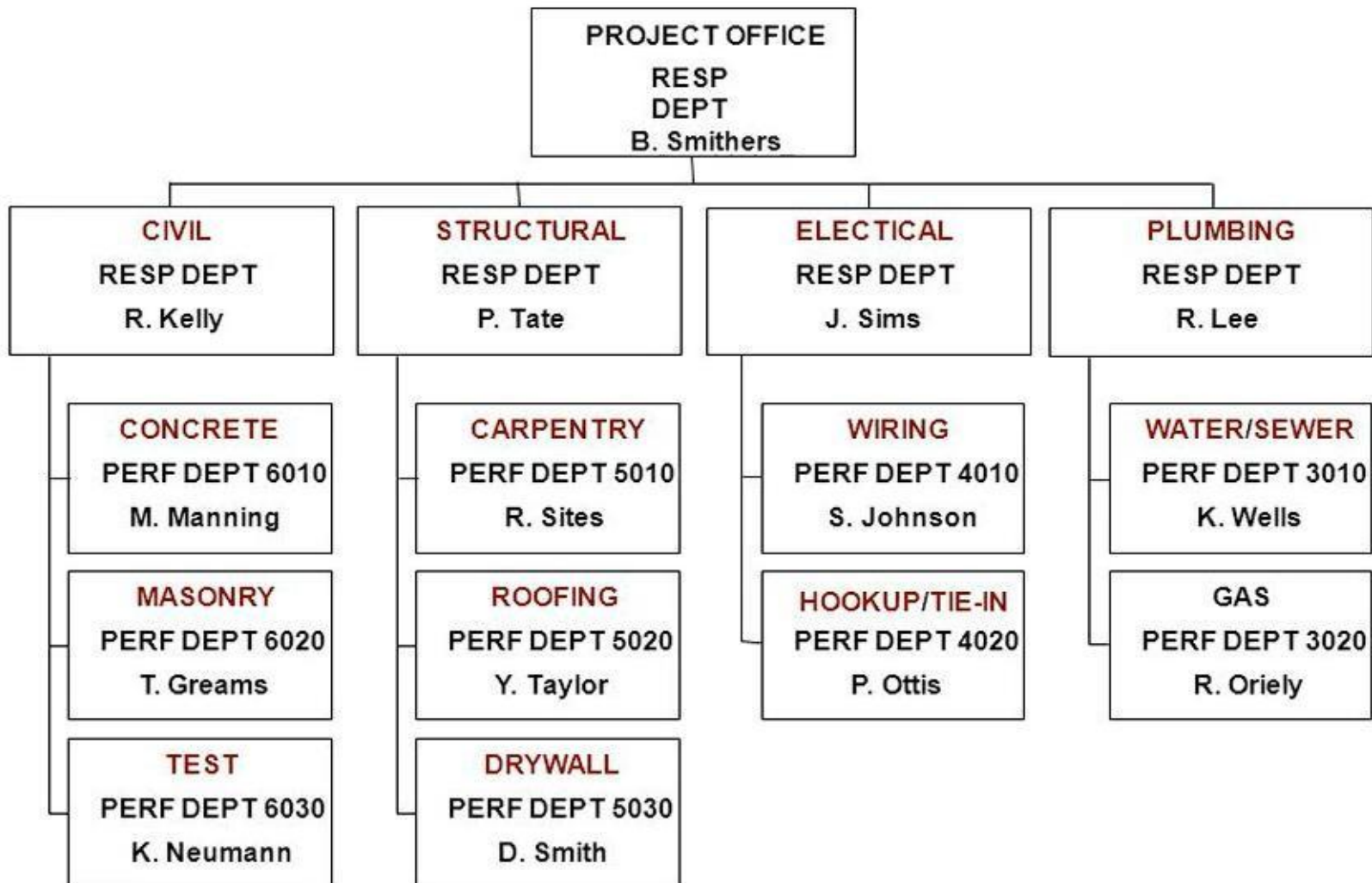
- Organization Breakdown Structure (OBS) can be defined as a framework or hierarchical representation of different roles within an organization, their responsibilities, and accountability. OBS can be implemented as a tool to manage the authorized work defined within the scope statement of the project.
- The OBS shares a deep relationship with WBS. OBS defines which entity would be responsible for the completion of a certain segment of WBS.
- This process is usually aided by the technique of Responsibility Assignment Matrix (RAM). This also helps to define the scope of work.

Organizational Breakdown Structure (OBS)

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Organization Breakdown Structure (OBS)



Responsibility Assignment Matrix

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(RAM)

- The technique of RAM is implemented in complex projects to assign different project members their respective activities or work packages.
- RAM is the visualization of the interaction between tasks that are needed to be completed for objectives and members of the project team. Different team members possess a different set of skills and they get assigned work based on their working capabilities or area of expertise.
- The figure next slide depicts a standard RAM, although there is room for variations based on variations in the project.

Responsibility Assignment Matrix (RAM)

2
2

	Jeff	Michael	Reto	YOU	Alex	Anna	Bill	Cindy	Felix	Fred	Hans	John	Livio	Luc	Marco	Paul	Peter	Sue	Ted	Tim
Planning / Schedule	R	A	I	C					C											Q
Risk Management		I	I	Q						A								R		
Quality Management			R	C						R										A
Procurement				R		Q				R								R		A
1. Specifications Listing								A		R								R		R
2. Site Requirements		C	A	R	Q						R									
3. Call for Tenders				Q	A	R	C				R							R		
4. Budget Approval				A	Q					R							R			R
5. Contract Negotiations			A		Q	R	R											R		

* R – Responsible (works on), A – Accountable, C – Consulted, I – Informed, Q – Quality Reviewer

Reference(s)

- † *Based in part on materials by Ms. Anam Rafique (Assistant Professor, Department of Computer Engineering in HITEC University Taxila Cantt).*
- † *Harold Kerzner, “Project Management, A System Approach to Planning, Scheduling & Controlling”, 10th Edition, John Wiley & Sons, Inc, 2012, ISBN: 978-0-470-27870-3.*
- † *Kathryn Bartol, David Martin, Margaret Tein, Graham Matthews “Management, A Pacific Rim Focus” 3rd Edition, McGraw-Hill Companies, Inc, 2002.*
- † *Gerry Johnson & kevan Scholes, “Exploring Corporate Strategy”, 8th Edition.*
- † *Related documents from open source, mainly internet.*